

1.7 Linear Independence

The homogeneous equations in Section 1.5 can be studied from a different perspective by writing them as vector equations. In this way, the focus shifts from the unknown solutions of $A\mathbf{x} = \mathbf{0}$ to the vectors that appear in the vector equations.

For instance, consider the equation

$$x_1 \begin{bmatrix} 1 \\ 2 \\ 3 \end{bmatrix} + x_2 \begin{bmatrix} 4 \\ 5 \\ 6 \end{bmatrix} + x_3 \begin{bmatrix} 2 \\ 1 \\ 0 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix} \quad (1)$$

This equation has a trivial solution, of course, where $x_1 = x_2 = x_3 = 0$. As in Section 1.5, the main issue is whether the trivial solution is the *only one*.

➤ LINEAR INDEPENDENCE

📖 **Definition:** An indexed set of vectors $\{\mathbf{v}_1, \dots, \mathbf{v}_p\}$ in $\mathbb{R}^n$ is said to be **linearly independent** if the vector equation

$$x_1 \mathbf{v}_1 + x_2 \mathbf{v}_2 + \dots + x_p \mathbf{v}_p = \mathbf{0}$$

has only the trivial solution. The set $\{\mathbf{v}_1, \dots, \mathbf{v}_p\}$ is said to be **linearly dependent** if there exist weights $c_1, \dots, c_p$, not all zero, such that

$$c_1 \mathbf{v}_1 + c_2 \mathbf{v}_2 + \dots + c_p \mathbf{v}_p = \mathbf{0} \quad (2)$$

❑ Equation (2) is called a **linear dependence relation** among $\mathbf{v}_1, \dots, \mathbf{v}_p$ when the weights are not all zero.

□ **Example 1:** Let $\mathbf{v}_1 = \begin{bmatrix} 1 \\ 2 \\ 3 \end{bmatrix}$, $\mathbf{v}_2 = \begin{bmatrix} 4 \\ 5 \\ 6 \end{bmatrix}$, and $\mathbf{v}_3 = \begin{bmatrix} 2 \\ 1 \\ 0 \end{bmatrix}$.

- a. Determine if the set $\{\mathbf{v}_1, \mathbf{v}_2, \mathbf{v}_3\}$ is linearly independent.
- b. If possible, find a linear dependence relation among $\mathbf{v}_1$, $\mathbf{v}_2$, and $\mathbf{v}_3$.

□ **Solution:** We must determine if there is a nontrivial solution of the equation on the previous slide.

- Row operations on the associated augmented matrix show that

$$\begin{bmatrix} 1 & 4 & 2 & 0 \\ 2 & 5 & 1 & 0 \\ 3 & 6 & 0 & 0 \end{bmatrix} \quad \begin{bmatrix} 1 & 4 & 2 & 0 \\ 0 & -3 & -3 & 0 \\ 0 & 0 & 0 & 0 \end{bmatrix}.$$

- x_1 and x_2 are basic variables, and x_3 is free.
- Each nonzero value of x_3 determines a nontrivial solution of (1).
- Hence, $\mathbf{v}_1, \mathbf{v}_2, \mathbf{v}_3$ are linearly dependent.

- b. To find a linear dependence relation among $\mathbf{v}_1$, $\mathbf{v}_2$, and $\mathbf{v}_3$, completely row reduce the augmented matrix and write the new system:

$$\begin{bmatrix} 1 & 0 & -2 & 0 \\ 0 & 1 & 1 & 0 \\ 0 & 0 & 0 & 0 \end{bmatrix} \quad \begin{array}{l} x_1 - 2x_3 = 0 \\ x_2 + x_3 = 0 \\ 0 = 0 \end{array}$$

- Thus, $x_1 = 2x_3$, $x_2 = -x_3$, and x_3 is free.
- Choose any nonzero value for x_3 —say, $x_3 = 5$.
- Then $x_1 = 10$ and $x_2 = -5$.

- Substitute these values into equation (1) and obtain the equation below.

$$10\mathbf{v}_1 - 5\mathbf{v}_2 + 5\mathbf{v}_3 = 0$$

- This is one (out of infinitely many) possible linear dependence relations among $\mathbf{v}_1$, $\mathbf{v}_2$, and $\mathbf{v}_3$.

- Suppose that we begin with a matrix $A = [a_1 \quad \cdots \quad a_n]$ instead of a set of vectors.
- The matrix equation $A\mathbf{x} = \mathbf{0}$ can be written as
$$x_1 a_1 + x_2 a_2 + \cdots + x_n a_n = \mathbf{0}.$$
- *Each linear dependence relation among the columns of A corresponds to a nontrivial solution of $A\mathbf{x} = \mathbf{0}$.*
- The columns of matrix A are linearly independent if and only if the equation $A\mathbf{x} = \mathbf{0}$ has *only the trivial solution*.

➤ SETS OF ONE VECTOR

- ❑ A set containing only one vector – say, $\mathbf{v}$ – is linearly independent if and only if $\mathbf{v}$ is not the zero vector.
- ❑ This is because the vector equation $x_1 \mathbf{v} = \mathbf{0}$ has only the trivial solution when $\mathbf{v} \neq \mathbf{0}$.
- ❑ The zero vector is linearly dependent because $x_1 \mathbf{0} = \mathbf{0}$ has many nontrivial solutions.

➤ SETS OF TWO VECTORS

- ❑ A set of two vectors $\{\mathbf{v}_1, \mathbf{v}_2\}$ is linearly dependent if at least one of the vectors is a multiple of the other.
- ❑ The set is linearly independent if and only if neither of the vectors is a multiple of the other.

➤ SETS OF TWO OR MORE VECTORS

THEOREM 7

Characterization of Linearly Dependent Sets

An indexed set $S = \{\mathbf{v}_1, \dots, \mathbf{v}_p\}$ of two or more vectors is linearly dependent if and only if at least one of the vectors in S is a linear combination of the others. In fact, if S is linearly dependent and $\mathbf{v}_1 \neq \mathbf{0}$, then some $\mathbf{v}_j$ (with $j > 1$) is a linear combination of the preceding vectors, $\mathbf{v}_1, \dots, \mathbf{v}_{j-1}$.

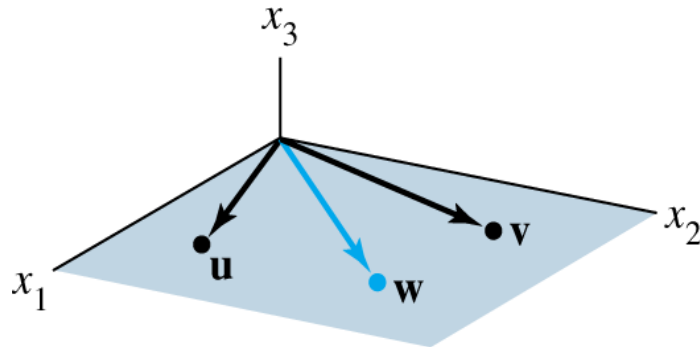
- ❑ Theorem 7 does *not* say that *every* vector in a linearly dependent set is a linear combination of the preceding vectors.
- ❑ A vector in a linearly dependent set may fail to be a linear combination of the other vectors.

❑ **Example 4:** Let $\mathbf{u} = \begin{bmatrix} 3 \\ 1 \\ 0 \end{bmatrix}$ and $\mathbf{v} = \begin{bmatrix} 1 \\ 6 \\ 0 \end{bmatrix}$. Describe the

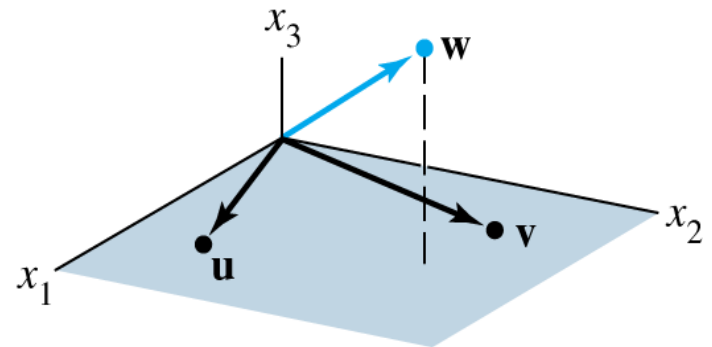
set spanned by $\mathbf{u}$ and $\mathbf{v}$, and explain why a vector $\mathbf{w}$ is in $\text{Span } \{\mathbf{u}, \mathbf{v}\}$ if and only if $\{\mathbf{u}, \mathbf{v}, \mathbf{w}\}$ is linearly dependent.

- ☐ **Solution:** The vectors $\mathbf{u}$ and $\mathbf{v}$ are linearly independent because neither vector is a multiple of the other, and so they span a plane in $\mathbb{R}^3$.
- ☐ Span $\{\mathbf{u}, \mathbf{v}\}$ is the x_1x_2 -plane (with $x_3 = 0$).
 - ☐ If $\mathbf{w}$ is a linear combination of $\mathbf{u}$ and $\mathbf{v}$, then $\{\mathbf{u}, \mathbf{v}, \mathbf{w}\}$ is linearly dependent, by Theorem 7.
 - ☐ Conversely, suppose that $\{\mathbf{u}, \mathbf{v}, \mathbf{w}\}$ is linearly dependent.
 - ☐ By theorem 7, some vector in $\{\mathbf{u}, \mathbf{v}, \mathbf{w}\}$ is a linear combination of the preceding vectors (since $\mathbf{u} \neq \mathbf{0}$).

So $\mathbf{w}$ is in $\text{Span}\{\mathbf{u}, \mathbf{v}\}$. Fig. 2 below



Linearly dependent,
 $\mathbf{w}$ in $\text{Span}\{\mathbf{u}, \mathbf{v}\}$



Linearly independent,
 $\mathbf{w}$ not in $\text{Span}\{\mathbf{u}, \mathbf{v}\}$

- Example 4 generalizes to any set $\{\mathbf{u}, \mathbf{v}, \mathbf{w}\}$ in $\mathbb{R}^3$ with $\mathbf{u}$ and $\mathbf{v}$ linearly independent.
- The set $\{\mathbf{u}, \mathbf{v}, \mathbf{w}\}$ will be linearly dependent if and only if $\mathbf{w}$ is in the plane spanned by $\mathbf{u}$ and $\mathbf{v}$.

THEOREM 9

If a set $S = \{v_1, \dots, v_p\}$ in $\mathbb{R}^n$ contains the zero vector, then the set is linearly dependent.

- **Proof:** By renumbering the vectors, we may suppose $v_1 = 0$.
- Then the equation $1v_1 + 0v_2 + \dots + 0v_p = 0$ shows that S is linearly dependent.

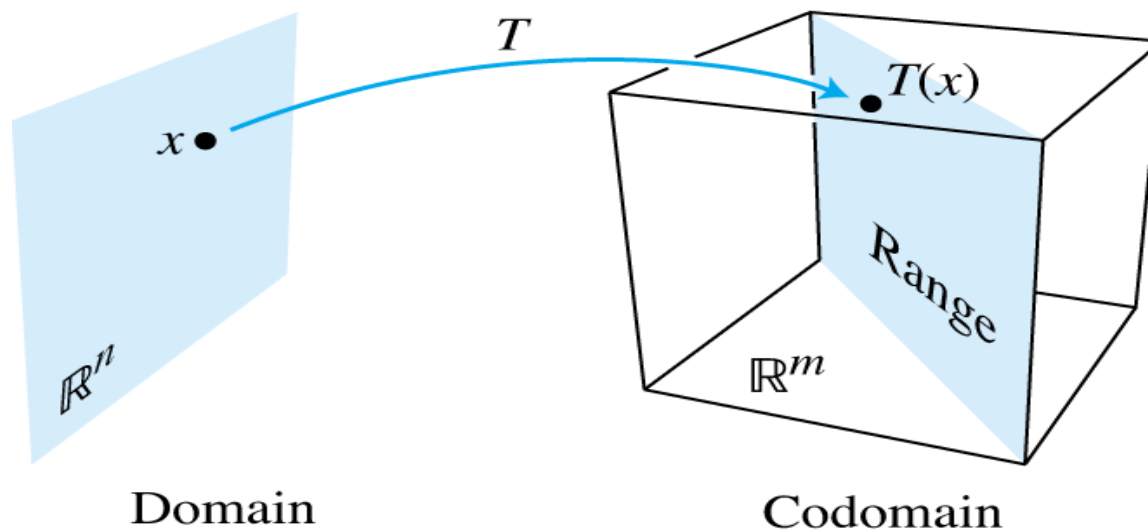
➤ LINEAR TRANSFORMATIONS

- ❑ (Section 1.8)
- ❑ A **transformation** (or **function** or **mapping**) T from $\mathbb{R}^n$ to $\mathbb{R}^m$ is a rule that assigns to each vector $\mathbf{x}$ in $\mathbb{R}^n$ a vector $T(\mathbf{x})$ in $\mathbb{R}^m$.
- ❑ The set $\mathbb{R}^n$ is called **domain of T** , and $\mathbb{R}^m$ is called the **codomain of T** .
- ❑ The notation $T: \mathbb{R}^n \rightarrow \mathbb{R}^m$ indicates that the domain of T is $\mathbb{R}^n$ and the codomain is $\mathbb{R}^m$.
- ❑ For $\mathbf{x}$ in $\mathbb{R}^n$, the vector $T(\mathbf{x})$ in $\mathbb{R}^m$ is called the **image of $\mathbf{x}$ (under the action of T)**.

TRANSFORMATIONS (變換)

range of

T



Domain, codomain, and range
of $T : \mathbb{R}^n \rightarrow \mathbb{R}^m$.

Example ---- MATRIX TRANSFORMATIONS (矩陣變換)

- ❑ For each $\mathbf{x}$ in $\mathbb{R}^n$, $T(\mathbf{x})$ is computed as $A\mathbf{x}$, where A is an $m \times n$ matrix.
- ❑ For simplicity, we denote such a *matrix transformation* by $\mathbf{x} \mapsto A\mathbf{x}$.
- ❑ Observe that the domain of T is $\mathbb{R}^n$ when A has n columns and the codomain of T is $\mathbb{R}^m$ when each column of A has m entries.
- ❑ The range of T is the set of all linear combinations of the columns of A , because each image $T(\mathbf{x})$ is of the form $A\mathbf{x}$.

For instance, the equations

$$\begin{matrix} & \uparrow A & & \uparrow \mathbf{x} & \uparrow \mathbf{b} & & \uparrow A & & \uparrow \mathbf{u} & \uparrow \mathbf{0} \\ \begin{bmatrix} 4 & -3 & 1 & 3 \\ 2 & 0 & 5 & 1 \end{bmatrix} & \begin{bmatrix} 1 \\ 1 \\ 1 \\ 1 \end{bmatrix} & = & \begin{bmatrix} 5 \\ 8 \end{bmatrix} & \text{and} & \begin{bmatrix} 4 & -3 & 1 & 3 \\ 2 & 0 & 5 & 1 \end{bmatrix} & \begin{bmatrix} 1 \\ 4 \\ -1 \\ 3 \end{bmatrix} & = & \begin{bmatrix} 0 \\ 0 \end{bmatrix} \end{matrix}$$

say that multiplication by A transforms $\mathbf{x}$ into $\mathbf{b}$ and transforms $\mathbf{u}$ into the zero vector. See Figure 1.

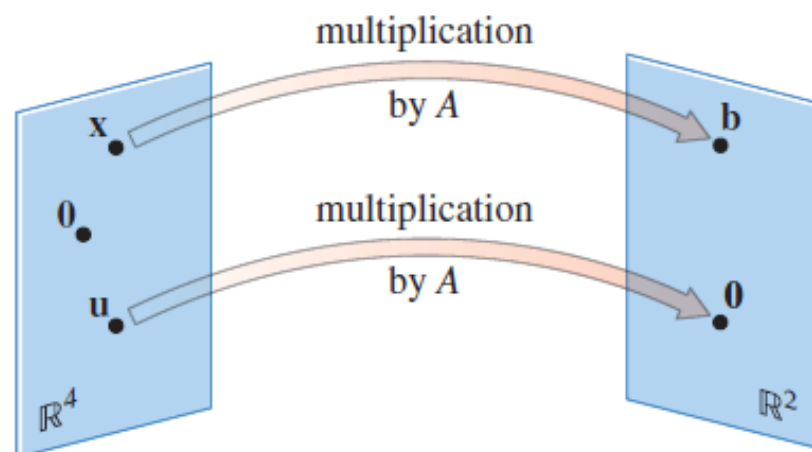


FIGURE 1 Transforming vectors via matrix multiplication.

MATRIX TRANSFORMATIONS

Example 1:

$$\text{Let } A = \begin{bmatrix} 1 & -3 \\ 3 & 5 \\ -1 & 7 \end{bmatrix}, \quad u = \begin{bmatrix} 2 \\ -1 \end{bmatrix}, \quad b = \begin{bmatrix} 3 \\ 2 \\ -5 \end{bmatrix}, \quad c = \begin{bmatrix} 3 \\ 2 \\ 5 \end{bmatrix}.$$

and define a transformation $T: \mathbb{R}^n \rightarrow \mathbb{R}^m$ by $T(\mathbf{x}) = A\mathbf{x}$, so that

$$T(\mathbf{x}) = A\mathbf{x} = \begin{bmatrix} 1 & -3 \\ 3 & 5 \\ -1 & 7 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} = \begin{bmatrix} x_1 - 3x_2 \\ 3x_1 + 5x_2 \\ -x_1 + 7x_2 \end{bmatrix}.$$

MATRIX TRANSFORMATIONS

- a. Find $T(\mathbf{u})$, the image of $\mathbf{u}$ under the transformation T .
- b. Find an $\mathbf{x}$ in $\mathbb{R}^2$ whose image under T is $\mathbf{b}$.
- c. Is there more than one $\mathbf{x}$ whose image under T is $\mathbf{b}$?
- d. Determine if $\mathbf{c}$ is in the range of the transformation T .

MATRIX TRANSFORMATIONS

Solution:

a. Compute

$$T(u) = Au = \begin{bmatrix} 1 & -3 \\ 3 & 5 \\ -1 & 7 \end{bmatrix} \begin{bmatrix} 2 \\ -1 \end{bmatrix} = \begin{bmatrix} 5 \\ 1 \\ -9 \end{bmatrix}.$$

b. Solve $T(x) = b$ for x . That is, solve $Ax = b$, or

$$\begin{bmatrix} 1 & -3 \\ 3 & 5 \\ -1 & 7 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} = \begin{bmatrix} 3 \\ 2 \\ -5 \end{bmatrix} \quad (1)$$

MATRIX TRANSFORMATIONS

- Row reduce the augmented matrix:

$$\begin{bmatrix} 1 & -3 & 3 \\ 3 & 5 & 2 \\ -1 & 7 & -5 \end{bmatrix} \rightarrow \begin{bmatrix} 1 & -3 & 3 \\ 0 & 14 & -7 \\ 0 & 4 & -2 \end{bmatrix} \rightarrow \begin{bmatrix} 1 & -3 & 3 \\ 0 & 1 & -.5 \\ 0 & 0 & 0 \end{bmatrix} \rightarrow \begin{bmatrix} 1 & 0 & 1.5 \\ 0 & 1 & -.5 \\ 0 & 0 & 0 \end{bmatrix} \quad (2)$$

- Hence $x_1 = 1.5$, $x_2 = -.5$, and $\mathbf{x} = \begin{bmatrix} 1.5 \\ -.5 \end{bmatrix}$.

- The image of this $\mathbf{x}$ under T is the given vector $\mathbf{b}$.

MATRIX TRANSFORMATIONS

- c. Any $\mathbf{x}$ whose image under T is $\mathbf{b}$ must satisfy equation (1).
 - From (2), it is clear that equation (1) has a unique solution.
 - So there is exactly one $\mathbf{x}$ whose image is $\mathbf{b}$.
- d. The vector $\mathbf{c}$ is in the range of T if $\mathbf{c}$ is the image of some $\mathbf{x}$ in $\mathbb{R}^2$, that is, if $\mathbf{c} = T(\mathbf{x})$ for some $\mathbf{x}$.
 - This is another way of asking if the system $A\mathbf{x} = \mathbf{c}$ is consistent.

MATRIX TRANSFORMATIONS

- To find the answer, row reduce the augmented matrix:

$$\begin{bmatrix} 1 & -3 & 3 \\ 3 & 5 & 2 \\ -1 & 7 & 5 \end{bmatrix} \quad \begin{bmatrix} 1 & -3 & 3 \\ 0 & 14 & -7 \\ 0 & 4 & 8 \end{bmatrix} \quad \begin{bmatrix} 1 & -3 & 3 \\ 0 & 1 & 2 \\ 0 & 14 & -7 \end{bmatrix} \quad \begin{bmatrix} 1 & -3 & 3 \\ 0 & 1 & 2 \\ 0 & 0 & -35 \end{bmatrix}$$

- The third equation, $0 = -35$, shows that the system is inconsistent.
- So $\mathbf{c}$ is *not* in the range of T .

Four Basic Questions about a transformation.

□ $T: \mathbb{R}^n \rightarrow \mathbb{R}^m$

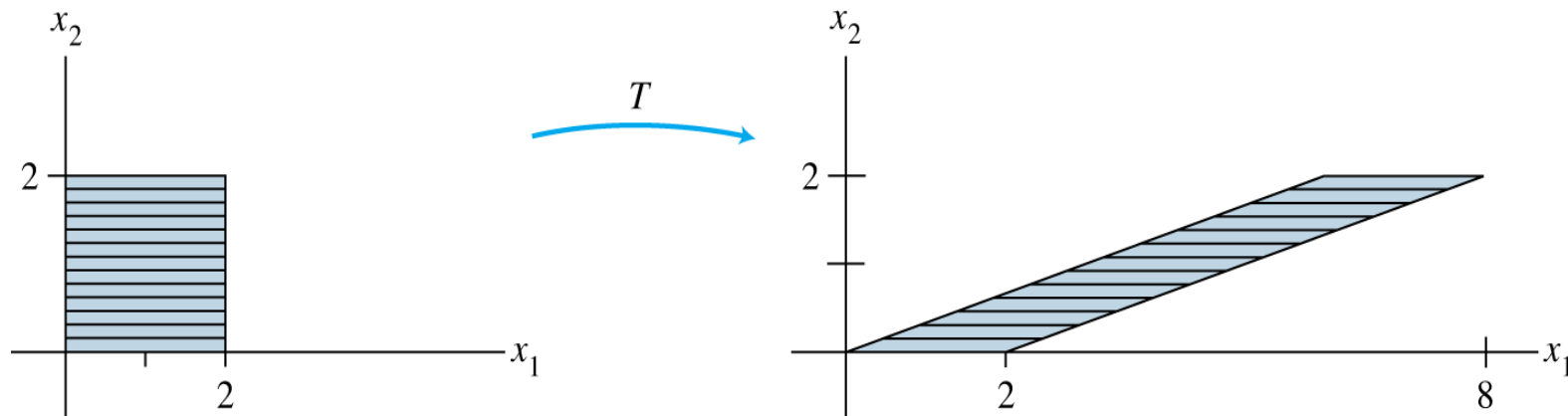
- Question 1: for any vector $\mathbf{u}$ in $\mathbb{R}^n$, what is the image of $\mathbf{u}$? (*function value*).
- Question 2: Find an $\mathbf{x}$ in $\mathbb{R}^n$ whose image under T is $\mathbf{b}$. (*independent value*)
- Question 3: Is there more than one $\mathbf{x}$ whose image under T is $\mathbf{b}$?
- Determine if $\mathbf{c}$ is in the range of the transformation T

(剪切變換) --- an example of matrix transformation

□ **Example 3:** Let $A = \begin{bmatrix} 1 & 3 \\ 0 & 1 \end{bmatrix}$. The transformation

$T: \mathbb{R}^2 \rightarrow \mathbb{R}^2$ defined by $T(\mathbf{x}) = A\mathbf{x}$ is called a **shear transformation**.

□ It can be shown that if T acts on each point in the 2×2 square shown belows, then the set of images forms the shaded parallelogram.



SHEAR TRANSFORMATION

- The key idea is to show that T maps line segments onto line segments and then to check that the corners of the square map onto the vertices of the parallelogram.

- For instance, the image of the point $u = \begin{bmatrix} 0 \\ 2 \end{bmatrix}$ is

$$T(u) = \begin{bmatrix} 1 & 3 \\ 0 & 1 \end{bmatrix} \begin{bmatrix} 0 \\ 2 \end{bmatrix} = \begin{bmatrix} 6 \\ 2 \end{bmatrix},$$

$$\begin{bmatrix} 2 \\ 2 \end{bmatrix} \quad \begin{bmatrix} 1 & 3 \\ 0 & 1 \end{bmatrix} \begin{bmatrix} 2 \\ 2 \end{bmatrix} = \begin{bmatrix} 8 \\ 2 \end{bmatrix}$$

LINEAR TRANSFORMATIONS

Definition: A transformation (or mapping) T is **linear** if:

- i. $T(\mathbf{u} + \mathbf{v}) = T(\mathbf{u}) + T(\mathbf{v})$ for all $\mathbf{u}, \mathbf{v}$ in the domain of T ;
- ii. $T(c\mathbf{u}) = cT(\mathbf{u})$ for all scalars c and all $\mathbf{u}$ in the domain of T .

Every matrix transformation is a linear transformation. Important examples of linear transformations that are not matrix transformations will be discussed in Chapters 4 and 5.

LINEAR TRANSFORMATIONS

- ❑ Linear transformations *preserve the operations of vector addition and scalar multiplication*.
- ❑ Property (i) says that the result $T(\mathbf{u} + \mathbf{v})$ of first adding $\mathbf{u}$ and $\mathbf{v}$ in $\mathbb{R}^n$ and then applying T is the same as first applying T to $\mathbf{u}$ and $\mathbf{v}$ and then adding $T(\mathbf{u})$ and $T(\mathbf{v})$ in $\mathbb{R}^m$.
- ❑ These two properties lead to the following useful facts

If T is a linear transformation, then

$$T(\mathbf{0}) = \mathbf{0} \quad (3)$$

and

$$T(c\mathbf{u} + d\mathbf{v}) = cT(\mathbf{u}) + dT(\mathbf{v}) \quad (4)$$

for all vectors $\mathbf{u}, \mathbf{v}$ in the domain of T and all scalars c, d .

LINEAR TRANSFORMATIONS

Property (3) follows from condition (ii) in the definition, because $T(0) = T(0u) = 0T(u) = 0$

- Property (4) requires both (i) and (ii):

$$T(cu + dv) = T(cu) + T(dv) = cT(u) + dT(v)$$

- *If a transformation satisfies (4) for all u, v and c, d , it must be linear.*
- (Set $c = d = 1$ for preservation of addition, and set for $d = 0$ preservation of scalar multiplication.)

LINEAR TRANSFORMATIONS

- Repeated application of (4) produces a useful generalization:

$$T(c_1 \mathbf{v}_1 + \dots + c_p \mathbf{v}_p) = c_1 T(\mathbf{v}_1) + \dots + c_p T(\mathbf{v}_p) \quad (5)$$

- In engineering and physics, (5) is referred to as a *superposition principle* (疊加原理) .

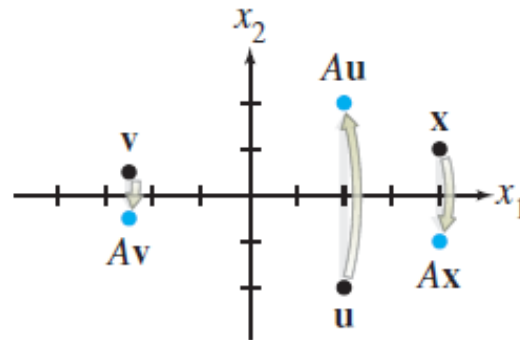
LINEAR TRANSFORMATIONS

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- Given a scalar r , define $T: \mathbb{R}^2 \rightarrow \mathbb{R}^2$ by $T(\mathbf{x}) = r\mathbf{x}$.
-
- T is called a **contraction** when $0 \leq r < 1$ and a **dilation** when $r > 1$. (收縮變換與拉伸變換)

Example: Given matrix of a matrix transformation, give the geometric

Let $A = \begin{bmatrix} 1 & 0 \\ 0 & -1 \end{bmatrix}$. Give a geometric description of the transformation $\mathbf{x} \mapsto A\mathbf{x}$.

Plot some random points (vectors) on graph paper to see what happens. A point such as $(4, 1)$ maps into $(4, -1)$. The transformation $\mathbf{x} \mapsto A\mathbf{x}$ reflects points through the x -axis (or x_1 -axis).



The transformation $\mathbf{x} \mapsto A\mathbf{x}$.

□ Question:

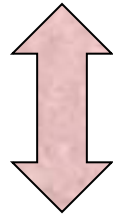
Given a geometric description of a linear transformation,

how to describe a linear transformation?

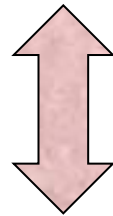
(how to give an explicit function relationship between x and y)

a “formula” for $T(\mathbf{x})$.

Every linear transformation from $\mathbb{R}^n$ to $\mathbb{R}^m$



A matrix transformation $\vec{x} \mapsto A\vec{x}$



the matrix A

➤ THE MATRIX OF A LINEAR TRANSFORMATION(Section 1.9)

□ **Theorem 10:** Let $T: \mathbb{R}^n \rightarrow \mathbb{R}^m$ be a linear transformation. Then there exists a unique matrix A such that

$$T(x) = Ax \text{ for all } x \text{ in } \mathbb{R}^n$$

□ In fact, A is the $m \times n$ matrix whose j^{th} column is the vector $T(e_j)$, where e_j is the j^{th} column of the identity matrix in $\mathbb{R}^n$

THE MATRIX OF A LINEAR TRANSFORMATION

▣ **Proof:** Write $x = I_n x = [e_1 \dots e_n]x = x_1 e_1 + \dots + x_n e_n$, and use the linearity of T to compute

$$T(x) = T(x_1 e_1 + \dots + x_n e_n) = x_1 T(e_1) + \dots + x_n T(e_n)$$

THE MATRIX OF A LINEAR TRANSFORMATION

- ❑ The matrix A in (3) is called the **standard matrix for the linear transformation T** .
- ❑ We know now that every linear transformation from $\mathbb{R}^n$ to $\mathbb{R}^m$ can be viewed as a matrix transformation, and vice versa. The term linear transformation focuses on a property of a mapping, while matrix transformation describes how such a mapping is implemented, as the example on the next slide illustrates.

THE MATRIX OF A LINEAR TRANSFORMATION

❏ **Example 2:** Find the standard matrix A for the dilation transformation $T(x) = 3x$, for x in $\mathbb{R}^2$.

❏ **Solution:** Write

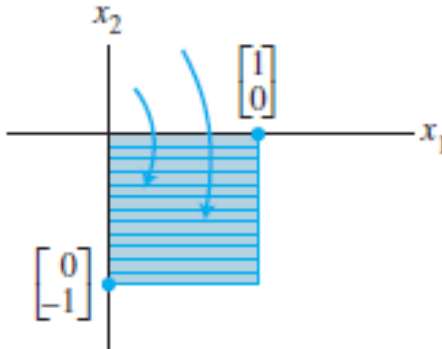
$$T(e_1) = 3e_1 = \begin{bmatrix} 3 \\ 0 \end{bmatrix} \quad \text{and} \quad T(e_2) = 3e_2 = \begin{bmatrix} 0 \\ 3 \end{bmatrix}$$

$$A = \begin{bmatrix} 3 & 0 \\ 0 & 3 \end{bmatrix}$$

GEOMETRIC LINEAR TRANSFORMATIONS OF $\mathbb{R}^2$

- Tables 1-4 illustrate other common geometric linear transformations of the plane.

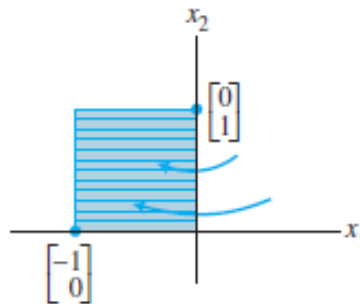
TABLE 1 Reflections

Transformation	Image of the Unit Square	Standard Matrix
Reflection through the x_1 -axis		$\begin{bmatrix} 1 & 0 \\ 0 & -1 \end{bmatrix}$

EXISTENCE AND UNIQUENESS QUESTIONS

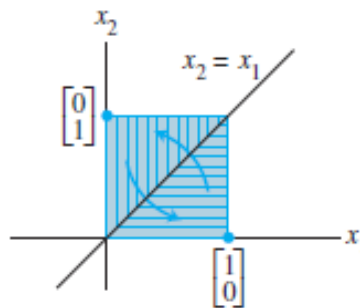
□ Table 1 continued:

Reflection through
the x_2 -axis



$$\begin{bmatrix} -1 & 0 \\ 0 & 1 \end{bmatrix}$$

Reflection through
the line $x_2 = x_1$

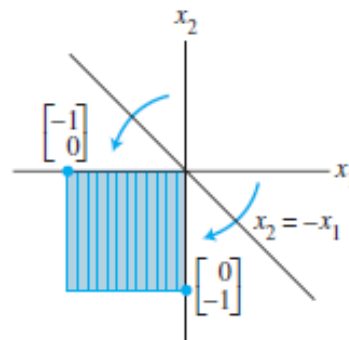


$$\begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix}$$

EXISTENCE AND UNIQUENESS QUESTIONS

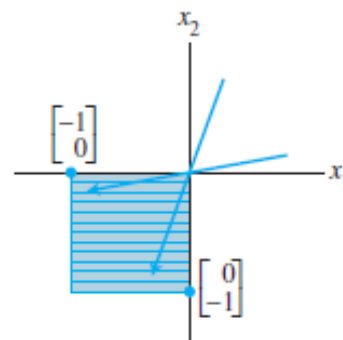
□ Table 1 continued:

Reflection through
the line $x_2 = -x_1$



$$\begin{bmatrix} 0 & -1 \\ -1 & 0 \end{bmatrix}$$

Reflection through
the origin



$$\begin{bmatrix} -1 & 0 \\ 0 & -1 \end{bmatrix}$$

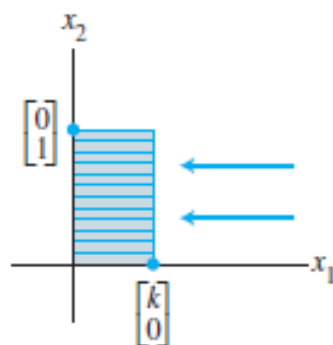
EXISTENCE AND UNIQUENESS

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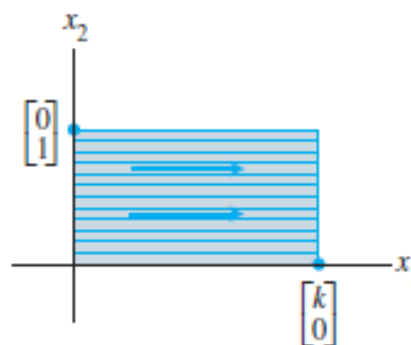
TABLE 2 Contractions and Expansions

Transformation	Image of the Unit Square	Standard Matrix
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Horizontal
contraction
and expansion



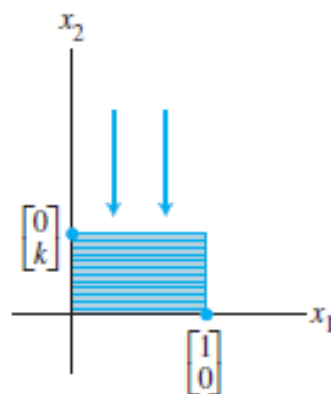
$$0 < k < 1$$



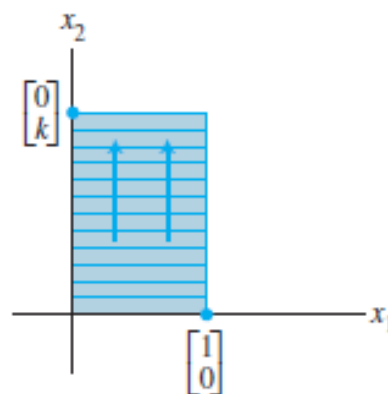
$$k > 1$$

$$\begin{bmatrix} k & 0 \\ 0 & 1 \end{bmatrix}$$

Vertical
contraction
and expansion



$$0 < k < 1$$



$$k > 1$$

$$\begin{bmatrix} 1 & 0 \\ 0 & k \end{bmatrix}$$

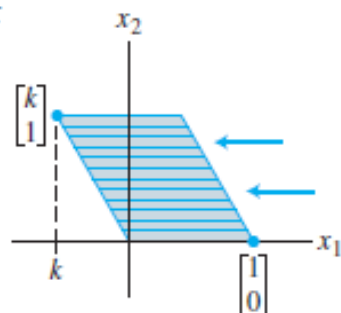
EXISTENCE AND UNIQUENESS

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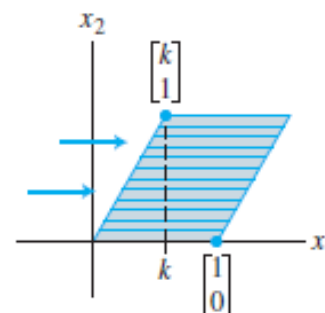
TABLE 3 Shears

Transformation	Image of the Unit Square	Standard Matrix
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Horizontal shear



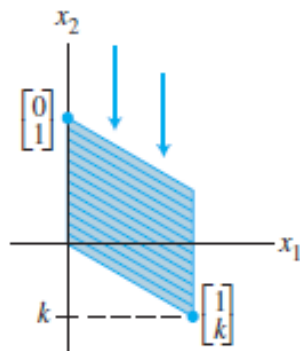
$k < 0$



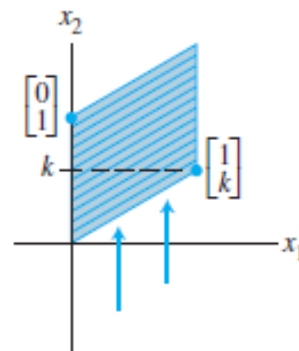
$k > 0$

$$\begin{bmatrix} 1 & k \\ 0 & 1 \end{bmatrix}$$

Vertical shear



$k < 0$



$k > 0$

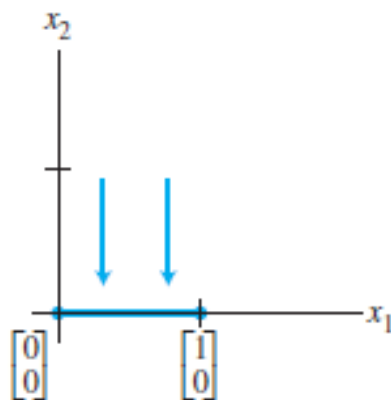
$$\begin{bmatrix} 1 & 0 \\ k & 1 \end{bmatrix}$$

EXISTENCE AND UNIQUENESS

TABLE 4 Projections

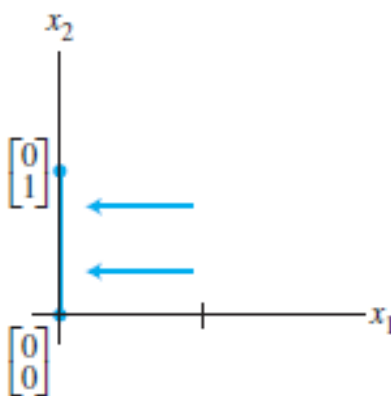
Transformation	Image of the Unit Square	Standard Matrix
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Projection onto
the x_1 -axis



$$\begin{bmatrix} 1 & 0 \\ 0 & 0 \end{bmatrix}$$

Projection onto
the x_2 -axis



$$\begin{bmatrix} 0 & 0 \\ 0 & 1 \end{bmatrix}$$

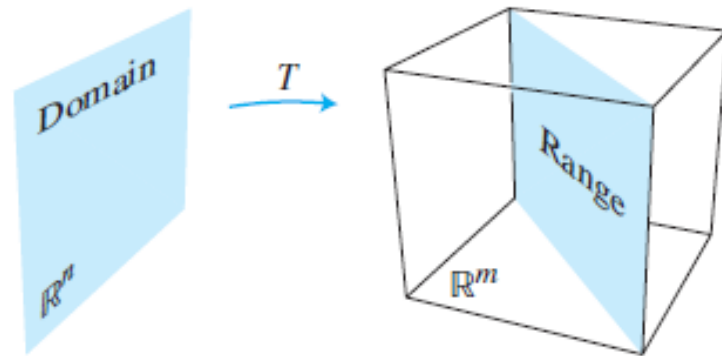
EXISTENCE AND UNIQUENESS

QUESTIONS

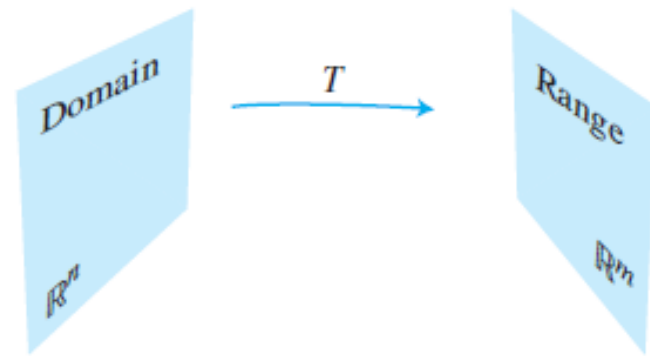
- **Definition:** A mapping $T: \mathbb{R}^n \rightarrow \mathbb{R}^m$ is said to be **onto** $\mathbb{R}^m$ if each $\mathbf{b}$ in $\mathbb{R}^m$ is the image of *at least one* $\mathbf{x}$ in $\mathbb{R}^n$.
- Equivalently, T is onto $\mathbb{R}^m$ when the range of T is all of the codomain $\mathbb{R}^m$. That is, T maps $\mathbb{R}^n$ onto $\mathbb{R}^m$ if, for each $\mathbf{b}$ in the codomain $\mathbb{R}^m$, there exists at least one solution of $T(\mathbf{x}) = \mathbf{b}$. “Does T map $\mathbb{R}^n$ onto $\mathbb{R}^m$?” is an existence question. The mapping T is not onto when there is some $\mathbf{b}$ in $\mathbb{R}^m$ for which the equation $T(\mathbf{x}) = \mathbf{b}$ has no solution. See the figure on the next slide.

EXISTENCE AND UNIQUENESS

QUESTIONS



T is not onto $\mathbb{R}^m$



T is onto $\mathbb{R}^m$

FIGURE 3 Is the range of T all of $\mathbb{R}^m$?

- Definition:** A mapping $T: \mathbb{R}^n \rightarrow \mathbb{R}^m$ is said to be **one-to-one** if each $\mathbf{b}$ in $\mathbb{R}^m$ is the image of *at most one* $\mathbf{x}$ in $\mathbb{R}^n$.

EXISTENCE AND UNIQUENESS

QUESTIONS

□ **Example 4:** Let T be the linear transformation whose standard matrix is

$$A = \begin{bmatrix} 1 & -4 & 8 & 1 \\ 0 & 2 & -1 & 3 \\ 0 & 0 & 0 & 5 \end{bmatrix}$$

□ Does T map $\mathbb{R}^4$ onto $\mathbb{R}^3$? Is T a one-to-one mapping?

EXISTENCE AND UNIQUENESS

QUESTIONS

- **Solution:** Since A happens to be in echelon form, we can see at once that A has a pivot position in each row. By Theorem 4 in Section 1.4, for each $\mathbf{b}$ in $\mathbb{R}^3$, the equation $A\mathbf{x}=\mathbf{b}$ is consistent. In other words, the linear transformation T maps $\mathbb{R}^4$ (its domain) onto $\mathbb{R}^3$.
- However, since the equation $A\mathbf{x}=\mathbf{b}$ has a free variable (because there are four variables and only three basic variables), each $\mathbf{b}$ is the image of more than one $\mathbf{x}$. This is, T is *not* one-to-one.

KERNEL AND RANGE OF A LINEAR TRANSFORMATION



Subspaces of vector spaces other than $\mathbb{R}^n$ are often described in terms of a linear transformation instead of a matrix.



Definition: A **linear transformation** T from a vector space V into a vector space W is a rule that assigns to each vector $\mathbf{x}$ in V a unique vector $T(\mathbf{x})$ in W , such that

- i. $T(\mathbf{u} + \mathbf{v}) = T(\mathbf{u}) + T(\mathbf{v})$ for all $\mathbf{u}, \mathbf{v}$ in V , and
- ii. $T(c\mathbf{u}) = cT(\mathbf{u})$ for all $\mathbf{u}$ in V and all scalars c .

Glossary

- | | |
|---------------------------|---------|
| 1. Square matrix | 方阵 |
| 2. (Row) Column Vector | (行)列向量 |
| 3. Parallelogram rule | 平行四边形法则 |
| 4. Scalar multiplication | 数乘 |
| 5. Linear combination | 线性组合 |
| 6. Parametric vector form | 参数向量形式 |
| 7. (Non)Homogeneous | (非)齐次的 |
| 8. (Non)Trivial solution | (非)平凡解 |